

# **Digital Voter Information & Management System (VIMS)**

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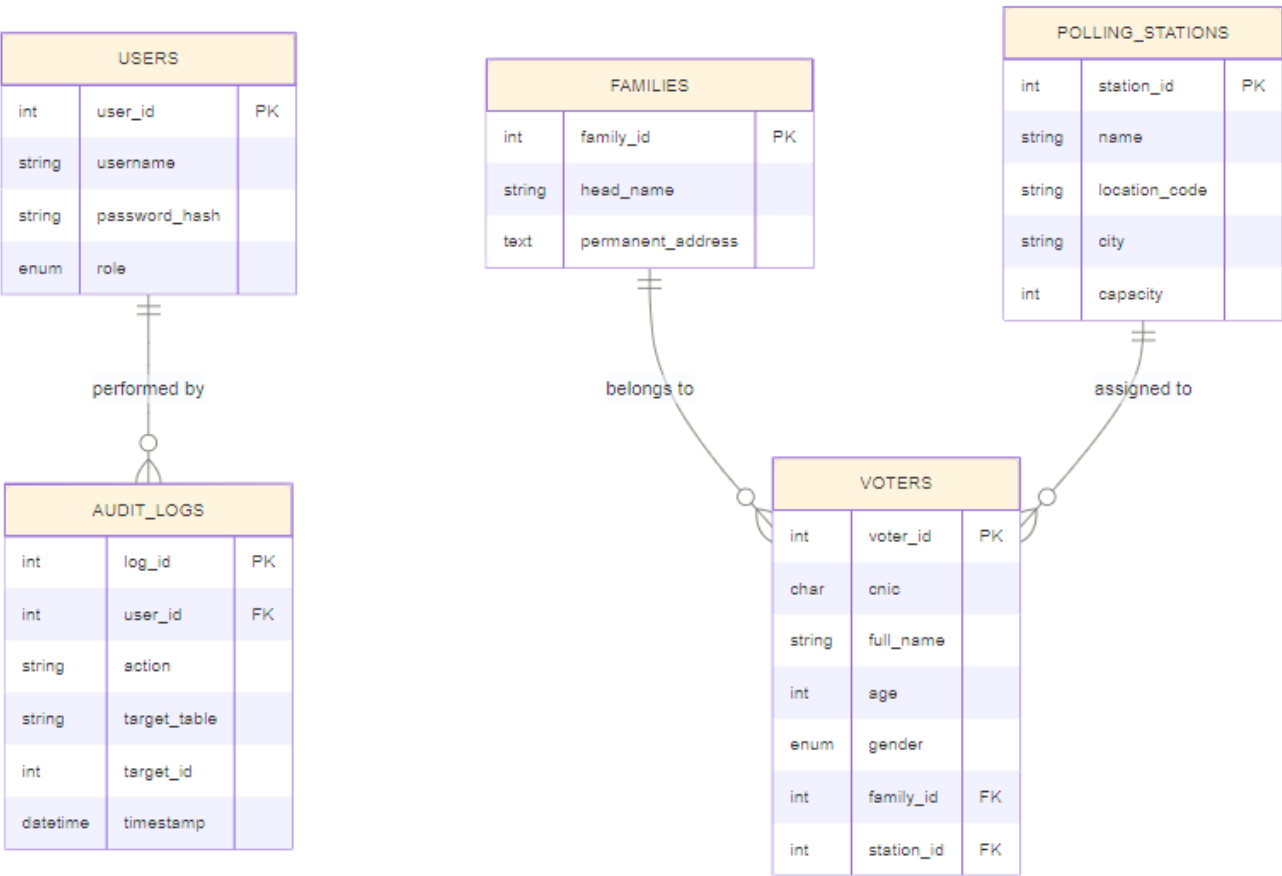
Submission Date: April 26, 2025

# Version Control Document

| Milestone                   | Version | Submission Date | Remarks             |
|-----------------------------|---------|-----------------|---------------------|
| ERD & Relational Schema     | v1.0    | April 26, 2025  | Changes Recommended |
| Add description with schema | V1.1    | May 6, 2025     |                     |

Milestone 1

Entity-Relationship Diagram (ERD)



Schema

TABLE: Users

Stores login credentials and role assignments for system personnel. Supports two access levels: Administrator (full CRUD access) and Viewer (read-only access). Passwords are stored as hashed values to ensure security.

`user_id` INT PRIMARY KEY, AUTO\_INCREMENT  
`username` VARCHAR(50) NOT NULL, UNIQUE  
`password_hash` VARCHAR(255) NOT NULL  
`role` ENUM NOT NULL -- 'Administrator' or 'Viewer'

### TABLE: PollingStations

*Represents individual polling stations where voters are assigned to cast their ballots. Each station has a unique location code, an associated city, and a defined voter capacity used for load monitoring and reporting.*

|               |              |                             |
|---------------|--------------|-----------------------------|
| station_id    | INT          | PRIMARY KEY, AUTO_INCREMENT |
| name          | VARCHAR(100) | NOT NULL                    |
| location_code | VARCHAR(20)  | NOT NULL, UNIQUE            |
| city          | VARCHAR(50)  | NOT NULL                    |
| capacity      | INT          | NOT NULL                    |

### TABLE: Families

*Represents family (Gharana) units that group multiple voters under a single household. Stores the head of the family and the permanent residential address, enabling family-centric views and filtering of the voter roll.*

|                   |              |                             |
|-------------------|--------------|-----------------------------|
| family_id         | INT          | PRIMARY KEY, AUTO_INCREMENT |
| head_name         | VARCHAR(100) | NOT NULL                    |
| permanent_address | TEXT         | NOT NULL                    |

### TABLE: Voters

*The central table of the system. Stores individual voter records including a unique CNIC, full name, age (enforced to be 18 or above), and gender. Each voter is relationally linked to their Gharana (family\_id) and their designated polling station (station\_id), forming the core of the system's relational integrity.*

|            |              |                                                              |
|------------|--------------|--------------------------------------------------------------|
| voter_id   | INT          | PRIMARY KEY, AUTO_INCREMENT                                  |
| cnic       | CHAR(15)     | NOT NULL, UNIQUE                                             |
| full_name  | VARCHAR(100) | NOT NULL                                                     |
| age        | TINYINT      | NOT NULL, CHECK (age >= 18)                                  |
| gender     | ENUM         | NOT NULL -- 'Male', 'Female', 'Other'                        |
| family_id  | INT          | FOREIGN KEY → Families(family_id) ON DELETE SET NULL         |
| station_id | INT          | FOREIGN KEY → PollingStations(station_id) ON DELETE RESTRICT |

### TABLE: AuditLogs

*Tracks all data-modification actions performed by administrators. Records which user made a change, the type of operation (INSERT, UPDATE, or DELETE), which table and record were affected, and the exact timestamp. Provides a full accountability trail for election data integrity.*

log\_id     INT     PRIMARY KEY, AUTO\_INCREMENT

user\_id    INT     NOT NULL, FOREIGN KEY → Users(user\_id) ON DELETE CASCADE

action     VARCHAR(20) NOT NULL -- 'INSERT', 'UPDATE', 'DELETE'

target\_table VARCHAR(50) NOT NULL

target\_id   INT     NOT NULL

timestamp   DATETIME   NOT NULL, DEFAULT NOW()